Highlights

Operating a gas-lifted well under severe slugging: time-averaged signatures from a two-fluid simulator, physics-informed surrogates of the choke and gas-lift decisions, and their operational optimisation

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- Gas-lift branch simulated with the two-fluid model, controlled by the platform gas pressure
- Warm-started Sobol' campaigns with adaptive horizon: 189 runs over choke and lift gas in a day
- Time-averaged markers at eight points replace trajectories as the training data
- Physics of the averages cuts the residuals six-fold and the error with few runs by two to four
- Optimum of revenue minus compression on the surrogate, with capacity constraints on predicted maxima

Operating a gas-lifted well under severe slugging: time-averaged signatures from a two-fluid simulator, physics-informed surrogates of the choke and gas-lift decisions, and their operational optimisation

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ABSTRACT

A gas-lifted satellite well with a flowline and a riser can produce in severe slugging, in a pressure cycle or in a steady state depending on two operating decisions, the opening of the topside choke and the pressure of the lift gas at the platform, and the time-averaged production of an unstable operating point is not what a steady-state model predicts for it. This paper builds the data and the surrogate models of those two decisions and uses them to optimise the operation of the well. The well, flowline, riser and gas-lift branch are simulated with the four-equation two-fluid model of a companion study, with the flow pattern identified dynamically and the gas-lift line, annulus and check valve simulated with the same model and controlled only by the platform gas pressure. Two sampling campaigns over the two decisions, in rounds of Sobol' points run as warm-started chains along the shortest path from the base case, with an adaptive horizon that detects the established regime and captures three cycles, produced 189 runs in a day on a laptop over 20 to 260 bar of lift gas; each run is reduced to time-averaged markers (mean, minimum, maximum and the two extreme frequencies) of pressure, void fraction and phase flow rates at eight characteristic points of the network. The map of the decisions shows a stability boundary that runs from 80 bar of lift gas with the choke nearly closed to 125 bar with it open, continuous injection modulated by the cycle below it, and a production gain of up to 34 % above it. Two surrogates of the mean liquid, gas and lift-gas rates and of their spans were trained on the markers: a neural network and a physics-informed network whose physics are the time-averaged mass balances of the well, the gas-lift branch and the valve, exact for the averages and evaluated at collocation points without simulation. With 123 training runs both networks predict the mean liquid rate with a test coefficient of determination of 0.90 and the gas and lift-gas rates with 0.996 to 0.999; the physics reduce the residuals of the balances four to six times, divide the error of the gas rates by four with ten training runs and by two with forty, predict the maxima of the liquid and lift-gas rates with 0.90 and 0.997, and leave the spans, the signature of slugging, as the quantity that neither network predicts well. The surrogate then enters an operational optimisation of the two decisions: the revenue of the mean liquid rate minus the cost of compressing the lift gas from 5 bar to the platform pressure, with the predicted maxima of the lift-gas and liquid rates bounded by the compressor and separator capacities. At the prices of the example the compression costs one per cent of the revenue and the optimum is interior, at a choke opening of 0.73 and 228 bar of lift gas, 6 % above the best point of the first campaign; a compressor capacity moves the optimum down the valve law, and the trade-off against the suction pressure is left for the next step.

1. Introduction

Continuous gas lift is the artificial-lift method of most deepwater satellite wells in Brazil, and its optimisation — how much gas to inject in each well of a field that shares a compressor — has a large literature built on steady-state performance curves (Souza, 2010). A well with a flowline and a riser, however, does not always have a steady state to be optimised: with the choke open and little gas it produces in severe slugging (Schmidt, Brill and Beggs, 1980; Taitel, 1986; Jansen, Shoham and Taitel, 1996), and the time-averaged production, the liquid surges that reach the separator and the pressure swings at the wellhead are quantities of the cycle, not of the unstable steady state that a

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nodal model would return for the same point. A companion study (Souza and Costa, 2026) developed a four-equation two-fluid model with dynamic flow-pattern identification that reproduces the cycle in a well–flowline–riser system of field dimensions, located the onset of slugging as the choke opens and showed, with a gas-lift branch added to the system, that injection at 100 to 120 bar regularises the cycle and from 130 bar suppresses it.

This paper is about the operation of that gas-lifted well. Its question is practical: given the two decisions an operator has, the opening of the topside choke and the pressure of the lift gas at the platform, what does the well deliver on average, with what surges and in which regime — and can a model cheap enough for optimisation answer that over the whole plane of the two decisions? The dynamic simulation is too expensive to be called inside an optimiser (minutes per point), so the paper builds a surrogate, and it does so with a method that follows the companion work on compositional steady-state surrogates (Carvalho, Costa, Nunes and Souza, 2026): a multilayer perceptron trained on simulated data, and the same network with a physics term (Raissi, Perdikaris and Karniadakis, 2019; Karniadakis, Kevrekidis, Lu, Perdikaris, Wang and Yang, 2021). The novelty is on both sides of the surrogate. On the data side, the targets are not profiles but time-averaged signatures of a dynamic simulation, and producing them in affordable time required a campaign design of its own: an adaptive horizon that recognises the established regime, warm starts along a shortest path through the sampled points, and a storage that keeps only what the training needs. On the physics side, the governing equations cannot be imposed on a time average of a cycling system; what can, exactly, are the time-averaged mass balances of the well, of the gas-lift branch and of the valve, and the paper shows that this much physics is what the data alone cannot supply.

Section 2 summarises the phenomenological model and the gas-lift branch. Section 3 describes the campaign, the horizon, the markers and the resulting map of the decisions. Section 4 presents the two surrogates and Section 5 compares them. Section 6 formulates the operational optimisation on the surrogate, with capacity constraints on the predicted maxima, and reports its first results.

2. The phenomenological model

The system is the well–flowline–riser of the companion study (Souza and Costa, 2026) with its gas-lift branch (Fig. 1): a vertical well of 1500 m (127 mm) fed by a linear inflow performance relation, a flowline of 2000 m (152.4 mm) whose last 500 m descend at 2° to a low point, a bend of 5 m radius, a riser of 317.5 m and 10 m of horizontal pipe to a choke that discharges into a separator at 10 bar. The gas-lift branch is a 4-in line that runs from the platform along the route of the production line to the wellhead, the casing–tubing annulus (9 5/8 in by 5 1/2 in) to a gas-lift valve 30 m above the bottom of the well, and a check valve fed by a chamber at the base of the annulus. The base well has a reservoir pressure of 200 bar, a productivity index of 5 $\mathrm{m^3\,d^{-1}\,bar^{-1}}$ and a gas–oil ratio of 50 $\mathrm{m^3\,m^{-3}}$; the fluid is the light oil of Carvalho (2022) frozen as a polytropic pair.

Every pipe, the gas pipes included, is described by the one-dimensional four-equation two-fluid model of the companion study: mass and momentum of each phase, closures by flow pattern (stratified, intermittent, bubbly, annular) selected dynamically on the local state through smooth logistic switches, a low-void rule, an objective virtual-mass term and an interfacial pressure in the slug unit that keep the model hyperbolic along the simulated trajectories. The staggered finite-volume discretisation with phase-wise donor cells and the conservative variable-step implicit integration of that study are kept unchanged, with a compiled right-hand side and exact Jacobians; the base mesh has 15 cells in the well, 35 in the flowline–riser pipe, 19 in the gas-lift line and 10 in the annulus, 323 states in all. The gas-lift branch is controlled by one number, the gas pressure P_{gas} at the platform end of the line: the injected rate is not prescribed but follows from P_{gas}, the static gas columns of the line and the annulus, the valve law $q = 0.0075\, C_V \sqrt{\rho_G\, \Delta P}$ ($C_V = 1.5$, ΔP in bar, regularised at the origin) and the pressure of the well at the injection point, so that it responds to whatever the well does; at 20 bar the valve never opens and the system is the base case of the companion study with the branch inert. The reservoir gas is the gas of the well; the lift gas adds to it at the injection point.

3. Data: campaign, horizon and markers

3.1. Sampling the two decisions

The inputs are the choke opening $a \in [0.1, 1]$ and the platform gas pressure P_{gas}, with the well fixed at its base state, in two campaigns: a first over 20 to 160 bar, and a second over 160 to 260 bar once the first had placed the optimum of Section 6 on its upper edge. Points are drawn from a scrambled Sobol' sequence (Sobol', 1967; Owen, 1998) in rounds of 25, so that the rounds of a campaign together remain a single space-filling design, and each round is

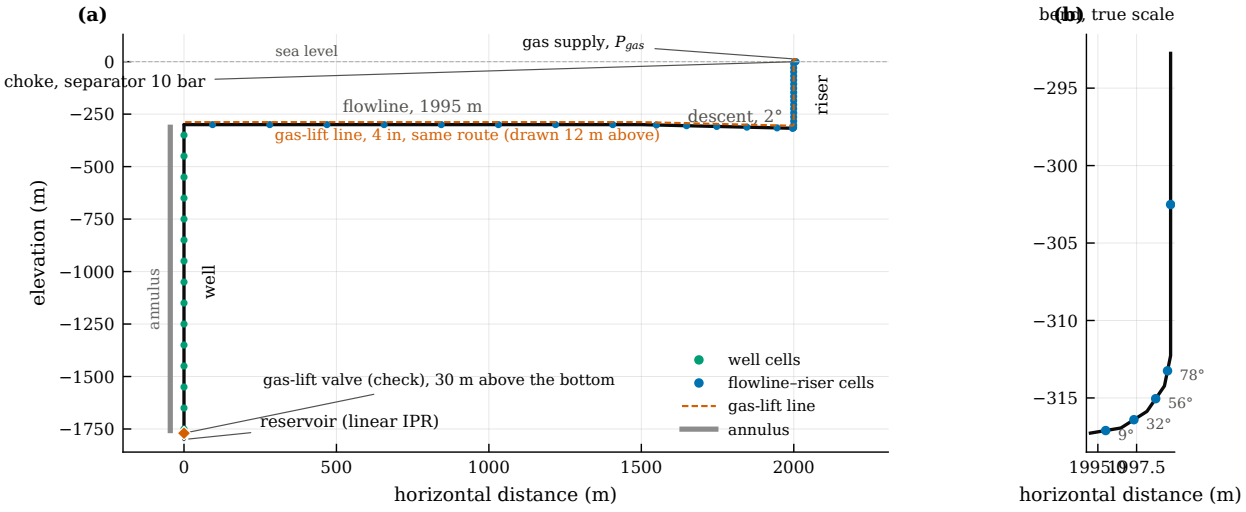


Figure 1: The well–flowline–riser system with the gas-lift branch (schematic, from the companion study): the gas-lift line follows the route of the production line, the annulus runs beside the well and the check valve injects 30 m above the bottom. The two operating decisions are the choke opening a and the platform gas pressure P_{gas}.

run as one chain: the first point is the one closest to the base case (choke open, 20 bar) and the run starts from the final state of the base run of the companion study (8000 s of severe slugging); every later point starts from the final state of the previous one, in the order of the shortest open path through the 25 points (nearest neighbour improved by 2-opt, on the inputs rescaled to $[-1, 1]$). The warm start spares the transient of a blocked start — three to five cycles in the companion study — and keeps the steps between consecutive runs small; a large drop of P_{gas} between neighbours is the one expensive transition, since the gas line and the annulus then drain towards the platform, and the shortest path avoids most of them. Each point is identified by a hash of its input tuple, and only what the training needs is stored: the markers below and the final state that seeds the next run.

3.2. Adaptive horizon with capture of cycles

Each run integrates in windows of 3000 s and monitors two signals, the pressure at the wellhead and the liquid rate through the choke. After a minimum time (4000 s from a blocked start, 1500 s from a warm start) the regime is taken as established when both signals are permanent (amplitude below 0.1 bar and 0.05 kg s^{-1}) or when two consecutive windows have the same envelope, that is, the same amplitude, mean, minimum and maximum within 10 % of the amplitude; the envelope criterion, rather than equality of periods, is what accepts the cycles whose blowouts alternate in size. The run then continues for three more cycles of the longest monitored period (600 s for a steady state), which are the capture window of the markers, and stops; a run that never settles stops at 24 000 s and is kept with the markers of its last 4000 s and a flag. A wall-clock budget of 40 min per run bounds the few points where the two horizontal cells between the riser and the choke switch flow pattern at every step, the one open point of the closures that the companion study reported; those runs are flagged and excluded from the training.

3.3. Markers at characteristic points

A run is reduced to markers of its capture window at eight points of the network: the bottom of the well, the injection point, the wellhead, the low point of the flowline, the top of the riser, the two ends of the gas-lift line and the base of the annulus. At each point four signals are recorded — pressure, void fraction, liquid and gas mass flow rates — and each signal is summarised by its mean, minimum, maximum and two frequencies, the inverse of the longest and of the shortest interval between successive maxima of the smoothed signal: equal for a periodic cycle, different when blowouts alternate, zero for a permanent signal. The gas injected through the valve is recorded in the same way, and a class is assigned from the wellhead pressure and the choke liquid rate as in the companion study (steady, pressure cycle or severe slugging when the peak liquid rate exceeds twice its mean). Trajectories are not kept: they would cost 12 MB per run against 10 kB of markers, and the surrogate consumes only the markers.

Table 1
The sampling campaign over the two decisions.

points (runs)	189
steady / pressure cycle / severe slugging	90 / 13 / 86
stop: steady / cycle captured / time limit / budget	83 / 91 / 2 / 13
simulated time per run (s): median, min–max	9600, 2030–24000
accepted steps per run: median, min–max	6213, 159–80038
integration time per run (min): median, max	3.8, 40.0

Base well of Part I ($P_{res} = 200\,\mathrm{bar}$, $J = 5\,\mathrm{m^3\,d^{-1}\,bar^{-1}}$, GOR 50) with the gas-lift branch; inputs sampled by scrambled Sobol' sequences in rounds of 25 over $a \in [0.1, 1]$ and $P_{gas} \in [20, 160]$ bar (first campaign) and [160, 260] bar (second campaign); each round starts from the final state of the base run at the point closest to it and continues along the shortest path with warm starts. Budget: 40 min of integration per run.

3.4. The map of the decisions

Table 1 summarises the two campaigns and Fig. 2 shows them on the plane of the decisions. Of the 189 runs, 86 produce in severe slugging, 13 in a pressure cycle and 90 in a steady state; 91 settled on a repeated cycle, 83 on a steady state, 2 reached the time limit and 13 the budget. The stability boundary is a diagonal that runs from about 80 bar of lift gas with the choke at 0.1 to about 125 bar with it open: the more open the choke, the more gas is needed to suppress the cycle. The check valve opens between 86 and 95 bar almost independently of the choke, so that below 90 bar the data are the base well with the choke as the only decision; between the opening of the valve and the boundary the injection is continuous and modulated by the cycle, which shortens and regularises; above the boundary the well is steady and its production rises with the gas, to $5.3\,\mathrm{kg\,s^{-1}}$ at 160 bar with the choke open, 34 % above the base case, and keeps rising, more slowly, in the second campaign, where every run that settled is steady. The peak liquid rate through the choke reaches $45\,\mathrm{kg\,s^{-1}}$ in the slugging region with the choke open and falls below $5\,\mathrm{kg\,s^{-1}}$ in the steady region. The thirteen runs that exhausted the budget sit where the riser receives the most gas — on the boundary at high opening in the first campaign, and in a strip at openings of 0.5 to 0.8 between 168 and 244 bar in the second — and are the pattern switching of the two horizontal cells at the top of the riser, which the class criterion reads as surges; they are excluded from the training and mark where the simulator, not the well, is least reliable.

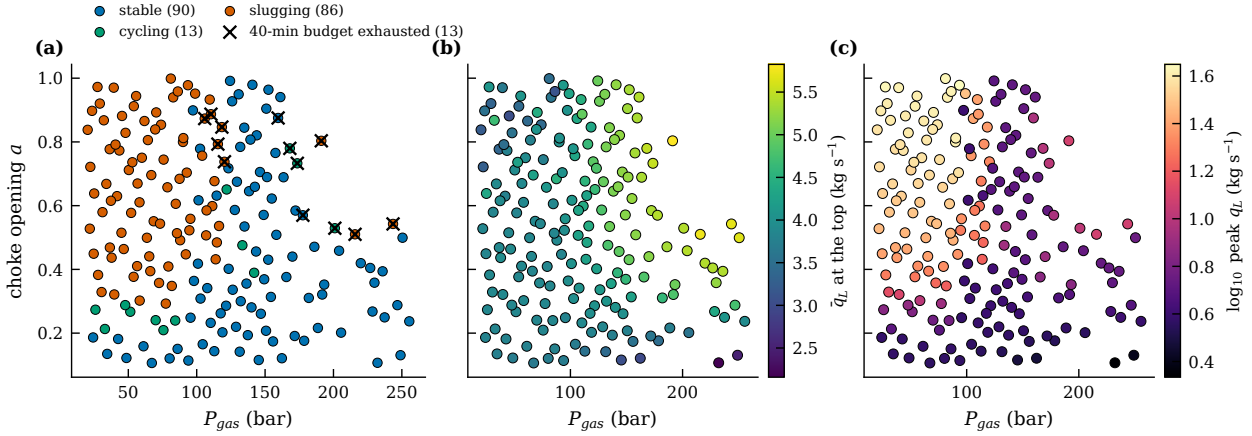


Figure 2: The 156 runs on the plane of the choke opening and the platform gas pressure: (a) class of the established regime, with the six runs that exhausted the 40-min integration budget before settling (crosses; their markers come from a partial trajectory and they are excluded from the training); (b) mean liquid rate through the choke; (c) peak liquid rate through the choke on a logarithmic scale.

4. Surrogates

4.1. Outputs and network

The surrogate maps the two decisions to eleven quantities of the capture window: the mean liquid and gas rates at the top of the riser, the mean gas-lift rate through the valve, the spans (maximum minus minimum) of the same three rates, which are the signature of slugging, the maxima of the liquid rate at the top and of the gas-lift rate, which the capacity constraints of the optimisation use, and three auxiliary mean pressures — at the bottom of the well, at the injection point and at the base of the annulus — that the physics of the next section needs. Spans and maxima are learnt as $\log(1 + x)$, so that they are never negative and their orders of magnitude are resolved. The network is a multilayer perceptron (Hornik, Stinchcombe and White, 1989) with two hidden layers of 32 hyperbolic-tangent units and a linear output, inputs and outputs standardised with the training statistics, trained with Adam (Kingma and Ba, 2015) in full batch for 4000 epochs on the mean squared error of the standardised outputs, with the epochs unrolled in a compiled scan so that a network trains in about a second; 70 % of the 176 retained runs train it and 30 % test it. A grid of one or two layers, 8 to 64 units and three learning rates (24 networks of each kind, Section 5) checked that the architecture is not critical.

4.2. Physics of the averages

A physics-informed network adds to the data loss the residual of the governing equations at collocation points that need not be simulated (Raissi et al., 2019). Here the targets are time averages of a dynamic system, and the time average of a cycle does not satisfy the two-fluid equations, which are nonlinear; what it satisfies exactly are the balances that are linear in the averaged quantities. Over a periodic cycle or a steady state, the mass of each phase in the system does not change on average, so that

$$R_1 = \bar{q}_L - J_L\,(P_{res} - \bar{P}_{bh}), \tag{1}$$

$$R_2 = \bar{q}_G - J_G\,(P_{res} - \bar{P}_{bh}) - \bar{q}_{gl}, \tag{2}$$

$$R_3 = \bar{P}_{an} - \kappa\,P_{gas}, \tag{3}$$

$$R_4 = \bar{q}_{gl} - 0.0075\,C_V\sqrt{\rho_G(\bar{P})\,\langle \bar{P}_{an} - \bar{P}_{inj}\rangle_+}, \tag{4}$$

where R_1 and R_2 are the liquid and gas balances between the linear inflow relation at the mean bottom-hole pressure and the mean rates at the top, R_3 is the static gas column from the platform to the base of the annulus ($\kappa = 1.138$ for this fluid and depth), and R_4 is the valve law written with the mean pressures, the only approximate relation of the four (the mean of a square root is not the square root of the mean, but the modulation of the valve pressure drop is within 25 % and the error is a few per cent), with a smooth gate that closes the valve when the mean annulus pressure falls below the mean injection pressure. The loss is the data term plus λ times the mean square of the four residuals, each scaled by the standard deviation of the quantity it compares, evaluated on 200 Latin-hypercube points (McKay, Beckman and Conover, 1979) of the plane of the decisions; $\lambda = 1$, and values from 0.1 to 100 give the same fit with residuals decreasing with λ. The residuals are closed-form functions of the network outputs, so that the gradient of the physics term is exact and cheap, unlike the detached-Jacobian construction that the compositional surrogates needed (Carvalho et al., 2026). The information in the physics is therefore not the dynamics but the accounting: the network is told that whatever enters the well on average leaves it on average, and that the lift gas is set by two mean pressures it also predicts.

5. Results

Table 2 and Fig. 3 compare the two networks on the 53 test runs. With the two campaigns together both networks predict the mean gas rate at the top and the mean gas-lift rate with R^2 above 0.996, the three mean pressures with 0.98 to 1.00, and the mean liquid rate with 0.90 (0.91 for the physics-informed network, with a root-mean-square error of $0.18\,\mathrm{kg\,s^{-1}}$): that rate varies little over the plane (2.2 to $5.8\,\mathrm{kg\,s^{-1}}$) and its time average over three cycles of a surging signal is the noisiest of the targets. With the first campaign alone (105 training runs) the data-only network reached only 0.57 on it and the physics raised it to 0.88; with the second campaign, whose steady points stretch the range of the rate, the data suffice and the physics show elsewhere: the scaled residuals of the four balances fall by a factor of four to six, and the balance R_1 is what keeps the liquid rate consistent with the bottom-hole pressure where the data are sparse. The maxima behave like the means they ride on: the maximum lift-gas rate reaches 0.997 with both networks,

Table 2
Test metrics of the two surrogates.

output	NN R^2	NN RMSE	PINN R^2	PINN RMSE
$\bar{q}_L$ at the top	0.902	0.180	0.908	0.175
$\bar{q}_G$ at the top	0.996	0.038	0.999	0.017
$\bar{q}_{gl}$ (gas lift)	0.996	0.036	0.999	0.015
span of q_L	0.391	11.616	0.630	9.056
span of q_G	0.257	0.502	-0.067	0.602
span of q_{gl}	0.486	0.040	0.431	0.042
max. q_L at the top	0.911	3.891	0.897	4.185
max. q_{gl}	0.997	0.032	0.997	0.029
$\bar{P}_{bh}$	0.978	1.945	0.979	1.899
$\bar{P}_{inj}$	0.978	1.942	0.982	1.745
$\bar{P}_{an}$	0.996	3.896	1.000	1.125
residuals R_1–R_4 (RMS, std units)	–	0.372, 0.010, 0.079, 0.067	–	0.064, 0.002, 0.015, 0.017

Test set of 53 runs (123 for training); RMSE in kg/s for the flow rates and bar for the pressures. Residuals: root-mean-square of the four scaled physics residuals on 400 held-out collocation points. Both networks have two hidden layers of 32 hyperbolic-tangent units; the PINN adds the physics term with $\lambda = 1$ on 200 Latin-hypercube collocation points.

since the injection is at most modulated by a quarter, and the maximum liquid rate, which carries the surge, 0.91 and 0.90. The spans are the weak point of both networks: the liquid span reaches 0.39 and 0.63; the gas span and the span of the injected gas stay below 0.5, since the amplitude of a surge in three cycles is a noisy marker by nature. Fig. 4 shows the physics-informed surrogate on the plane of the decisions: the mean liquid rate rises with the lift gas above the boundary and falls as the choke closes, the lift-gas rate is zero below 90 bar and rises with P_{gas} as the valve law dictates, and the maximum liquid rate marks the slugging region, with surges of 20 to 40 kg s^{-1} below the boundary against 5 above it, and a narrow band of artefacts along the boundary, where the data change class within a few bar.

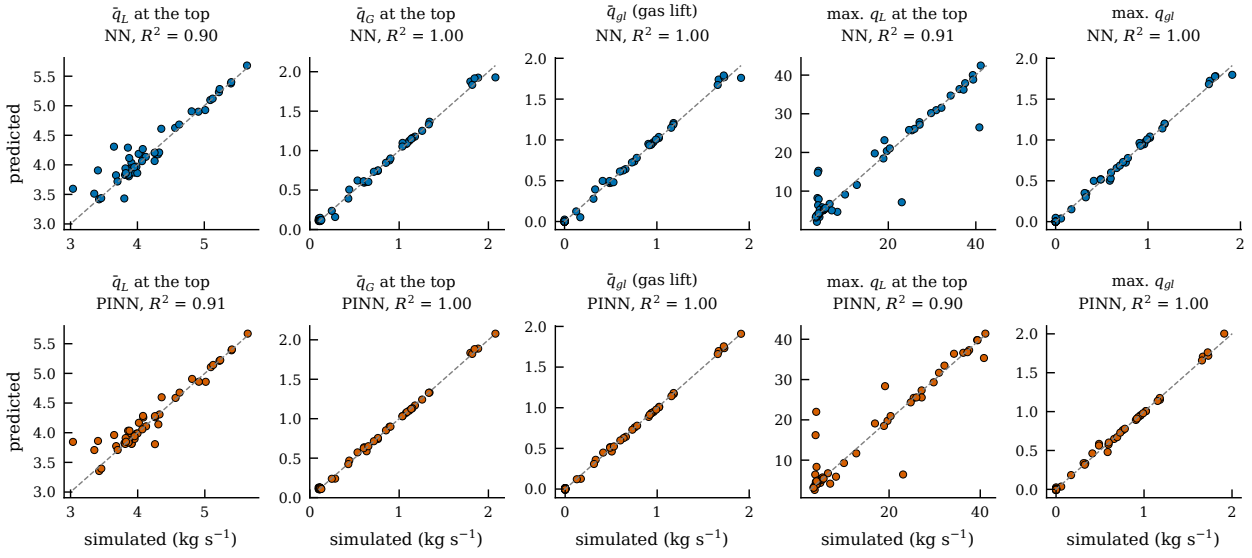


Figure 3: Test parity of the mean liquid, gas and lift-gas rates and of the maxima of the liquid and lift-gas rates, for the data-only network (top row) and the physics-informed network (bottom row).

With few data the physics counts more. Table 3 and Fig. 5 retrain both networks on 10, 20 and 40 runs drawn from the training set, three draws each, and test them on the same 45 runs. The physics-informed network has the lower error for the gas and lift-gas rates at every size — a quarter of the error of the data-only network with 10 runs, a third with 20 and a half with 40 — and for the liquid rate at every size, from 0.36 to 0.20 kg s^{-1} against 0.38 to 0.30; with 40 runs it is within 15 % of what either network reaches with 123. The hyperparameter grid confirms that the architecture is not what limits the fit: of 24 architectures, three data-only and six physics-informed networks reach a mean R^2 of

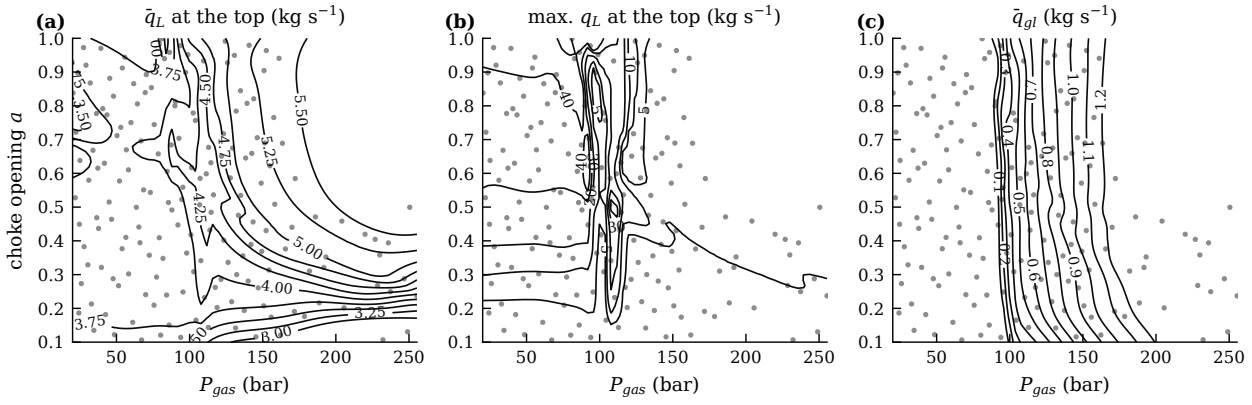


Figure 4: The physics-informed surrogate on the plane of the decisions as labelled level curves, with the positions of the simulated runs (grey dots): (a) mean liquid rate at the top; (b) maximum liquid rate at the top, the surge that reaches the separator; (c) mean gas-lift rate.

Table 3
Sparse-data study: test error of the mean rates against the number of training runs.

training points	$\bar{q}_L$ NN	$\bar{q}_L$ PINN	$\bar{q}_G$ NN	$\bar{q}_G$ PINN	$\bar{q}_{gl}$ NN	$\bar{q}_{gl}$ PINN
10	0.378 ± 0.033	0.357 ± 0.025	0.196 ± 0.083	0.044 ± 0.006	0.199 ± 0.079	0.034 ± 0.005
20	0.355 ± 0.163	0.268 ± 0.063	0.116 ± 0.012	0.042 ± 0.012	0.104 ± 0.017	0.038 ± 0.010
40	0.300 ± 0.075	0.202 ± 0.021	0.058 ± 0.008	0.028 ± 0.003	0.056 ± 0.006	0.025 ± 0.003
123	0.180	0.175	0.038	0.017	0.036	0.015

Test RMSE (kg/s), mean ± standard deviation over three random subsets of the training set; last row: the full training set.

0.8 over the six requested outputs, and the differences between architectures are smaller than the difference the physics makes with few data.

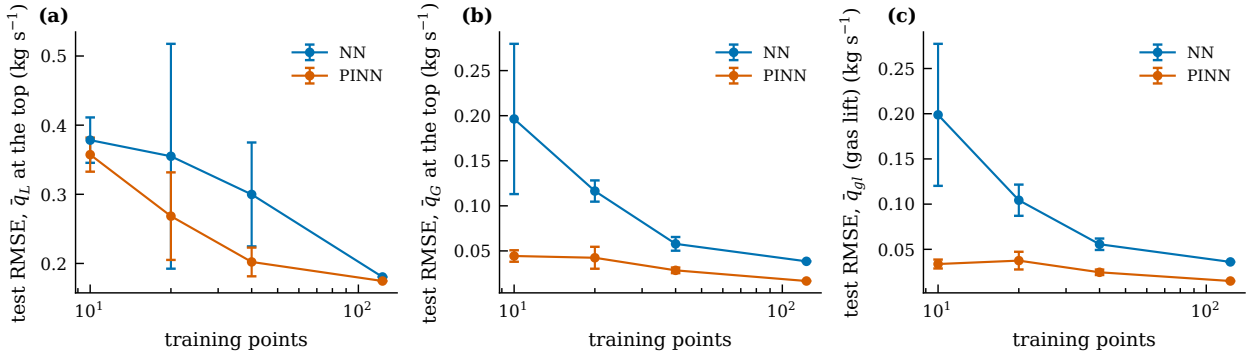


Figure 5: Learning curves of the two surrogates: test root-mean-square error of (a) the mean liquid rate, (b) the mean gas rate and (c) the mean gas-lift rate against the number of training runs (mean and standard deviation over three draws; the last point is the full training set).

6. Operational optimisation

6.1. Formulation

The operator chooses the choke opening a and the platform gas pressure P_{gas} to maximise the daily margin of the well,

$$F(a, P_{gas}) = c_{oil}\, Q_o(\bar{q}_L) - c_e\, 24\, W(\bar{q}_{gl}, P_{gas}), \tag{5}$$

the revenue of the oil minus the cost of compressing the lift gas, subject to the capacities of the compressor and of the separator,

$$\max q_{gl} \le Q_{gl}^{cap}, \qquad \max q_L \le Q_L^{cap}, \tag{6}$$

written on the maxima of the rates over the cycle, which the surrogate predicts (Section 4.1), and not on their means: a compressor and a separator are sized for peaks. The oil rate Q_o follows from the mean liquid rate with the liquid mass per stock-tank cubic metre of the base fluid (700 kg m^{-3}, oil and solution gas); the compression power W is that of a polytropic compression of the mean lift-gas rate from the suction pressure (5 bar) to P_{gas} in stages of ratio at most four with intercooling to the suction temperature ($k = 1.3$, 313 K, efficiency 0.75, molar mass 19.5 g mol^{-1} from the gas density of the fluid). All the quantities in Eqs. (5) and (6) come from the physics-informed surrogate, so that F and the constraints are smooth closed-form functions of the two decisions, evaluated in microseconds. The problem is solved by enumeration on a 201×201 grid of the plane restricted to the feasible nodes, followed by a sequential quadratic programming refinement with the two constraints and the bounds, and the optimum is confirmed by one run of the simulator at that point.

6.2. First results

[Seed of the optimisation study; the Pareto front against the suction pressure and the real capacities are pending.] Fig. 6 shows the objective on the plane of the decisions for an oil price of 70 \$ per barrel and an energy price of 0.10 \$ per kW h, with the runs of the campaigns by class. On the first campaign alone the optimum was the corner of maximum gas and open choke, (1, 160 bar), with a margin of 290×10^3 \$/d, and the simulator run at that point, started from the base state, confirmed the surrogate to 2 % (a mean liquid rate of 5.40 against 5.38 kg s^{-1}, a lift-gas rate of 1.17 against 1.15 and a bottom-hole pressure of 67.0 against 67.2 bar); an optimum on the edge of the sampled box says only that the box is too small, which is why the second campaign was run. With both campaigns the optimum is interior: $(a, P_{gas}) = (0.73, 228\,\text{bar})$, with a margin of 309×10^3 \$/d, a revenue of 313×10^3 \$/d against a compression cost of 3.3×10^3 \$/d (1384 kW for 1.75 kg s^{-1} of lift gas), 6 % above the corner of the first campaign. The compression is one per cent of the revenue, so the objective is the production, which the surrogate puts at a maximum of 5.75 kg s^{-1} where the gain of the lift gas is balanced by the friction of a tubing and riser that carry 1.75 kg of gas per second, and which falls on both sides: the choke partly closed holds back the gas, the choke open and more gas cost production. The simulator run at this optimum is pending; the optimum sits beside the strip of budget-limited runs of Section 3.4, which is where the confirmation matters most. The optimum leaves the box only when the energy costs about fifty times more (5 \$ per kW h) or the oil a third as much (20 \$ per barrel), and then it jumps to the well without gas lift at an opening of 0.65, the best point of the slugging region, since the margin of the lift gas no longer pays its compression anywhere.

The constraints change the picture as soon as they bind. With a compressor of 0.8 kg s^{-1} and a separator that accepts peaks of 20 kg s^{-1}, the compressor constraint is active and the optimum moves to (1.0, 117 bar) with a margin of 268×10^3 \$/d: the valve law makes the lift-gas rate a function of P_{gas} alone, so a compressor capacity fixes the platform pressure, and the choke opens fully because at that pressure the production still rises with the opening. That point lies on the stability boundary, with a predicted peak liquid rate of 18.5 kg s^{-1} against the separator limit of 20: the separator constraint is a hair from active, and with a smaller separator it is the constraint that keeps a gas-limited optimum away from severe slugging.

7. Conclusions

A dynamic two-fluid simulation of a gas-lifted well can feed a surrogate of its operating decisions once the campaign is designed for it: warm-started chains along the shortest path through the sampled points, an adaptive horizon that recognises the established regime and captures three cycles, and markers at characteristic points instead of trajectories. Over the plane of the choke opening and the platform gas pressure the well has a stability boundary from 80 to 125 bar of lift gas, continuous injection modulated by the cycle below it and up to 34 % more production above it at 160 bar. Of the two surrogates trained on the 189 runs, the physics-informed one, whose physics are the time-averaged mass balances of the well and of the gas-lift branch, predicts the mean liquid rate with R^2 0.91, the gas and lift-gas rates with 0.999 and their maxima with 0.90 and 0.997, satisfies the balances four to six times better than the data-only network, needs a quarter to a half of the data for the same error, and leaves the spans of the rates as the quantity to be handled by

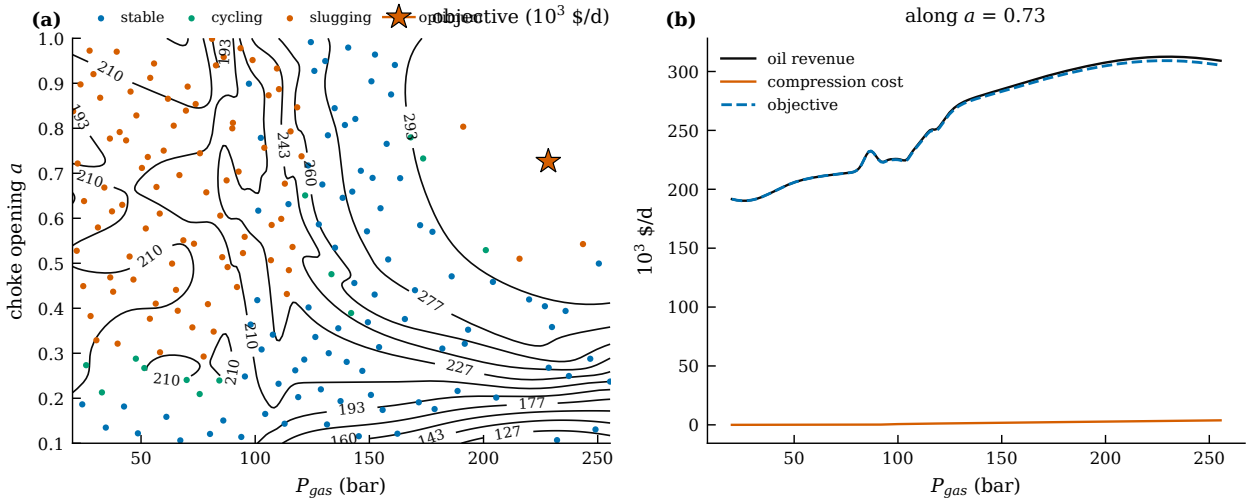


Figure 6: Operational optimisation on the surrogate at 70 $ per barrel and 0.10 $ per kW h: (a) objective of Eq. (5) on the plane of the decisions (level curves, 10^3 $/d), the runs of the campaign by class and the unconstrained optimum (star); (b) revenue, compression cost and objective along the platform gas pressure with the choke at the optimal opening.

the class and the stability map rather than by regression. On this surrogate the optimisation of the revenue minus the compression cost, with the predicted maxima bounded by the compressor and separator capacities, is a two-variable problem solved by enumeration and refined in a fraction of a second: at the prices of the example the compression is one per cent of the revenue and the optimum is interior, at an opening of 0.73 and 228 bar, where the gain of the lift gas is balanced by the friction it costs; a compressor capacity makes the valve law the active constraint and moves the optimum down the platform pressure to the stability boundary. The trade-off against the suction pressure, as a Pareto front, is the next step.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used AI-assisted tools for language editing and code assistance. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Data availability

The simulator used in this study is not publicly available; the campaign design (sampling, horizon, markers) and the surrogates are described in full so that they can be reproduced with an independent implementation, and the table of markers and the trained networks are available from the corresponding author upon reasonable request.

References

Carvalho, G.A.d., 2022. Modelagem e simulação composicional do escoamento multifásico em tubulações com geração de modelos substitutos. Dissertação de mestrado. Universidade do Estado do Rio de Janeiro, Instituto de Química. Rio de Janeiro.

Carvalho, G.A.d., Costa, A.L.H., Nunes, G.C., Souza, J.N.M.d., 2026. Steady-state compositional gas–liquid flow in production pipelines. Part II: neural and physics-informed surrogates with the axial position as a feature. Companion paper, submitted to Geoenergy Science and Engineering.

Hornik, K., Stinchcombe, M., White, H., 1989. Multilayer feedforward networks are universal approximators. Neural Netw. 2, 359–366. doi:10.1016/0893-6080(89)90020-8.

Jansen, F.E., Shoham, O., Taitel, Y., 1996. The elimination of severe slugging—experiments and modeling. Int. J. Multiph. Flow 22, 1055–1072. doi:10.1016/0301-9322(96)00027-4.

Karniadakis, G.E., Kevrekidis, I.G., Lu, L., Perdikaris, P., Wang, S., Yang, L., 2021. Physics-informed machine learning. Nat. Rev. Phys. 3, 422–440. doi:10.1038/s42254-021-00314-5.

Kingma, D.P., Ba, J., 2015. Adam: A method for stochastic optimization, in: 3rd International Conference on Learning Representations (ICLR), San Diego, CA. ArXiv:1412.6980.

McKay, M.D., Beckman, R.J., Conover, W.J., 1979. A comparison of three methods for selecting values of input variables in the analysis of output from a computer code. Technometrics 21, 239–245. doi:10.1080/00401706.1979.10489755.

Owen, A.B., 1998. Scrambling Sobol' and Niederreiter–Xing points. J. Complexity 14, 466–489. doi:10.1006/jcom.1998.0487.

Raissi, M., Perdikaris, P., Karniadakis, G.E., 2019. Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. J. Comput. Phys. 378, 686–707. doi:10.1016/j.jcp.2018.10.045.

Schmidt, Z., Brill, J.P., Beggs, H.D., 1980. Experimental study of severe slugging in a two-phase-flow pipeline–riser pipe system. Soc. Pet. Eng. J. 20, 407–414. doi:10.2118/8306-PA.

Sobol', I.M., 1967. On the distribution of points in a cube and the approximate evaluation of integrals. USSR Comput. Math. Math. Phys. 7, 86–112. doi:10.1016/0041-5553(67)90144-9.

Souza, J.N.M., 2010. Modelagem e simulação de escoamento multifásico em dutos de produção de óleo e gás natural. D.Sc. thesis. Escola de Química, Universidade Federal do Rio de Janeiro. Rio de Janeiro, Brazil. In Portuguese.

Souza, J.N.M.d., Costa, A.L.H., 2026. Severe slugging with a four-equation two-fluid model and dynamic flow-pattern identification: well-posedness, discretisation, time integration and choke stability maps. Companion paper (Part I of this study), submitted.

Taitel, Y., 1986. Stability of severe slugging. Int. J. Multiph. Flow 12, 203–217. doi:10.1016/0301-9322(86)90026-1.